

CRISTOPHE CHEN

cristophe.chen@uwaterloo.ca | tophchen.com | linkedin.com/in/CristopheChen | github.com/tophuu

EDUCATION

University of Waterloo

Bachelor of Computer Science (Honours), Co-op

Waterloo, ON

Sept 2024 – May 2028

SKILLS

Languages: Python, Java, Kotlin, JavaScript, TypeScript, C/C++, C#, Ruby, Go, SQL, HTML/CSS

Technologies: React, React Native, Tailwind CSS, Next.js, Node.js, Express.js, FastAPI, Temporal, Flask, Three.js, PostgreSQL, MongoDB, DynamoDB, Redis, RabbitMQ, Kafka, GraphQL, gRPC, Drizzle, Sequelize, Pandas, NumPy

Tools: Git, Docker, Kubernetes, Supabase, Firebase, AWS, GCP, Terraform

EXPERIENCE

Wealthsimple

Sept 2026 – Dec 2026

Software Engineer Intern

Toronto, ON

- Building **Kotlin** ingestion pipelines and **Temporal** corporate action workflows for CAD-denominated options, helping scale the monthly active trader base from **24k** to **50k**.
- Migrating corporate action processing from 2 parallel publishing paths (Oracle ledger + real-time balance) to a unified financial-event processing platform using **Temporal** and **Kafka**, ensuring deterministic ledger adjustments for **millions** of client accounts.

Atlassian

May 2026 – Aug 2026

Software Engineer Intern

Vancouver, BC

- Increased Loom video discoverability by **9x** for search and agentic contexts by engineering a permission-alignment ingestion pipeline in **TypeScript** and **Node.js** to synchronize **OpenSearch** and **Atlassian Teamwork Graph**.
- Built a **RabbitMQ** worker for an entity-agnostic bulk-processing framework with dry-run, rollback, and full audit logging, enabling reliable large-scale workloads across both **PostgreSQL** and **TiDB**.
- Enabled the bulk-operation platform to backfill access and discoverability policies across **390M** existing videos by implementing task leasing and checkpointed keyset pagination.
- Delivered an admin interface for automating public-link expiration policies across **3,000+** workspaces using **React**, **GraphQL**, and workspace-scoped **RBAC**.
- Reduced data retrieval latency by up to **54%** during a **63 PiB** zero-downtime migration from **AWS S3** to a **cloud-agnostic** object store by developing dual-write storage handlers and dynamic AsyncLocalStorage routing across **108** shard models.

Qolytics

May 2025 – Aug 2025

Software Engineer Intern

Calgary, AB

- Engineered a high-throughput ingestion pipeline to finalize **thousands** of events per session into compressed telemetry objects, reducing storage payload size by **60%** and accelerating downstream processing.
- Shipped a **React** and **Three.js** dashboard for **80+** enterprise clients by dynamically mapping raw wearable telemetry to 3D skeletal nodes for motion reconstruction and **Plotly** analysis.
- Accelerated biomechanical insight generation by **45%** by designing efficient **FastAPI** endpoints and optimizing **SQLAlchemy** queries.
- Provisioned identity infrastructure via **AWS Cognito**, implementing robust **OAuth 2.0/OIDC** authentication flows to secure access for **1,000+** active users.

Cohere

Jan 2025 – Apr 2025

Data Engineer

Toronto, ON

- Optimized **RLHF** pipelines and built training datasets for LLMs, helping reach **86.2%** on the MBPP-Plus benchmark and **51.5%** on the LBPP benchmark.
- Reviewed over **2,000** coding and STEM entries to evaluate the quality and accuracy of LLM-generated responses.

PROJECTS

Data Undersampling Visualizer | ThreeJS, NextJS, Flask, Scikit-learn, Numpy, TypeScript

- Awarded **Best DEI AI Hack** out of **620+** participants for building a bias-mitigation tool for model training.
- Engineered a data pipeline that embedded raw data into **384-dimensional embeddings**, subsequently utilizing **k-means clustering methods** to segment data for bias analysis.
- Automated dataset outlier detection by integrating **LLM inference** to classify and isolate clustered attributes

Multimodal AI Tutor | NextJS, Flask, Redis, Vapi

- Developed a multimodal AI-tutor synchronizing real-time screen analysis, live sketching, and concurrent voice streams.
- Slashed system latency by **80%** (8-15s to 2-3s) by implementing Redis-backed semantic caching.
- Orchestrated text-to-speech streams via **VAPI SDK** to guarantee real-time conversational responsiveness.